Carolina Veiga

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Profile

With a background in Environmental Engineering and a Ph.D. in Computing, I integrate environmental sciences, visual analytics, and data provenance to tackle urban challenges. My doctoral thesis focused on visualization systems for atmospheric modeling, particularly extreme weather events. My research focuses on three main areas: (1) developing visualization systems that help domain experts effectively interpret complex, uncertain, and high-dimensional data; (2) integrating AI to support expert-in-the-loop analysis; and (3) enhancing data visualization education by leveraging AI to foster analytical, creative, and critical thinking skills. Cross-domain collaborations have led to publications in leading computer science journals and conferences, including one award-winning work. As a professor, I strive to connect theory with practice, engage students in real-world visualization challenges, and inspire creative, data-informed, and collaborative solutions.

Education

2018–2023 **Ph.D. in Computing** — Niterói, Brazil
Universidade Federal Fluminense
Advisor: Dr. Marcos Lage/Co-advisor: Dr. Marcio Cataldi
Thesis: Visual management and analysis of weather forecasts ensembles
Supported by CAPES

2013–2016 **M.S. in Byosystems Engineering** — Niterói, Brazil
Universidade Federal Fluminense
Advisor: Dr. Marcio Cataldi
Thesis: Numerical evaluation of the influence of urbanization in the convection and precipitation patterns in the Metropolitan Region of São Paulo
Supported by CAPES

2008–2013 **B.S. in Environmental Engineering** — Niterói, Brazil
Universidade Federal Fluminense
Advisor: Dr. Mônica da Hora
Thesis: Ecological flow estimation in a stretch of the Paracatu river using the wetted perimeter method; Thesis 2: Ecological flow estimation in a stretch of the Piabanha river using the wetted perimeter method

Professional Experience

2025–Present **Teaching/Research Postdoctoral Fellow** — Harrisonburg, VA
James Madison University
Supervisor: Dr. Vetria Byrd
Conducting a systematic literature review on data visualization pedagogies and literacy; contributing to an NSF proposal; observing classes; delivered lectures and labs, supervised a workshop, and preparing to co-teach in Spring.

2023–2025 **Postdoctoral Researcher Associate** — Chicago, IL, US
Discovery Partners Institute, part of University of Illinois System
Supervisor: Dr. Ashish Sharma
Contributed to multiple research projects, developing three visualization systems and co-authoring two IEEE VIS papers; also participated in AGU and AMS conferences

2021 **Researcher of Research and Development Project** — Niterói, RJ, Brazil
AMBMET CONSULTORIA LTDA for Casa dos Ventos Comercializadora de Energia S.A, with co-participation of the Universidade Federal Fluminense
Research Project: Study to improve short- and medium-term discharge forecasting

2018 **Researcher of Research and Development Project** Niterói, RJ, Brazil
AMBMET CONSULTORIA LTDA for Valora Energia, with co-participation of the Universidade Federal Fluminense
Research Project: Implementation of a conceptual rainfall-discharge model for scenario forecasting

2015-2016 **Environmental Engineer** Rio de Janeiro, RJ, Brazil
Instituto Estadual do Ambiente (State Institution of the Environment)
Research Project: Development of a discharge prognostic system prototype by ensemble in quick response basins in support of prevention of natural disasters
Supported by CNPq

2013 **Member of Extension Project** Niterói, RJ, Brazil
Universidade Federal Fluminense
Research Project: Diagnosis of basic sanitation conditions of Oriximiná City (PA), aiming support to the elaboration of sanitation plans, programs, and projects; Elaboration of the draught bills of the Municipal Policy on Basic Sanitation and Solid Waste and the Municipal Plan of Basic Sanitation

2012-2013 **Research project member as an undergraduate student** Niterói, RJ, Brazil
Universidade Federal Fluminense
Research Project: Development of a weather numerical prediction system by high spatial resolution ensemble to the Niterói City (RJ)

2009-2011 **Engineer Intern** Rio de Janeiro, RJ, Brazil
Eletrobras Furnas
Development of the Annual Greenhouse Gas (GHG) Inventory in 2009, 2010 and 2011, including the study of methodologies, correction, and improvement of the excel spreadsheets, presentation and training given; Development of the computerized system for collecting company data needed for the GHG Inventory.

Main Publications

J: journal, C: conference/symposium, W: workshop, A: abstract. Note: VIS conference papers appear in a special issue of the IEEE TVCG journal

[C] 2025 Urbanite: A Dataflow-Based Framework for Human-AI Interactive Alignment in Urban Visual Analytics
G. Moreira, L. Ferreira, **C. Veiga**, M. Hosseini, F. Miranda
IEEE Transactions on Visualization and Computer Graphics 2025 (Accepted, to appear)

[A] 2025 Processing and Transforming Environmental Justice Data: Multiple Approaches and Urban Field Campaign Collaboration
A. Jain, **C. Veiga**, A. Sharma
American Meteorological Society 2025

[A] 2025 e-JUST: Environmental Justice using Urban Scalable Toolkit: Development and Enhancements
C. Veiga, P. Li, F. Miranda, G. Moreira, S. Wu, K. Jogi, A. Gautam, M. Colombo, S. Tokkar, E. Makra, A. Sharma
American Meteorological Society 2025

[C] 2025 Curio: A Dataflow-Based Framework for Collaborative Urban Visual Analytics
G. Moreira, M. Hosseini, **C. Veiga**, L. Alexandre, N. Colaninno, D. Oliveira, N. Ferreira, M. Lage, F. Miranda
IEEE Transactions on Visualization and Computer Graphics, v. 31(1), p. 1224-1234 (IEEE VIS 2024)

[W] 2024 Urban computing for climate and environmental justice: early perspectives from two research initiatives
C. Veiga, A. Sharma, D. Oliveira, M. Lage, F. Miranda
IEEE VIS 2024 Workshop on Visualization for Climate Action and Sustainability

[J] 2024	Heavy rainfall event in Nova Friburgo (Brazil): numerical sensitivity analysis using different parameterization combinations in the WRF model **C. Veiga**, M. G. A. J. da Silva, F. P. da Silva *Natural Hazards, v. 120, p. 11641-11664*
[A] 2024	e-JUST: Environmental Justice using Urban Scalable Toolkit A. Sharma, **C. Veiga**, P. Li, F. Miranda, G. Moreira, S. Wu, M. Budhathoki, A. Tiwari, J. Wei, M. Turk, E. Makra *American Meteorological Society 2024*
[C] 2024	ProWis: a visual approach for building, managing, and analyzing weather simulation ensembles at runtime **C. V. F. de Souza**, S. M. Bonnet, D. de Oliveira, M. Cataldi, F. Miranda, M. Lage *IEEE Transactions on Visualization and Computer Graphics, v.30(1), p.738-747* Best Paper Honorable Mention (IEEE VIS 2023)
[C] 2023	Division problems: interpretation, solution and planning in Lesson Study M. A. V. F. de Souza, **C. V. F. de Souza** *XVI Internamerican Conference on Mathematics Education*
[J] 2022	Visualizing simulation ensembles of extreme weather events **C. V. F. de Souza**, P. C. L. Barcellos, L. Crissaff, M. Cataldi, F. Miranda, M. Lage *Computer & Graphics, v. 104, p. 162-172*
[J] 2022	A comparative study of methods for the visualization of probability distributions of geographical data S. Srabanti, **C. V. F. de Souza**, E. Silva, M. Lage, N. Ferreira, F. Miranda *Multimodal Technologies and Interaction, v. 6(7), p. 53*
[J] 2021	Finantial education and sustainable development: a systematic literature review R. Leffler, **C. V. F. de Souza**, M. A. V. F. de Souza *Jornal Internacional de Estudos em Educação Matemática, v. 14(4), p. 502-513*
[J] 2017	Numerical evaluation of the influence of urbanization in the convection and precipitation patterns in the Metropolitan Region of São Paulo R. Leffler, **C. V. F. de Souza**, R. H. O. Rangel, M. Cataldi *Revista Brasileira de Meteorologia, v. 32(4), p. 495-508*

Awards

2023	**IEEE VIS 2023 Best Honorable Mention** For “ProWis: a visual approach for building, managing, and analyzing weather simulation ensembles at runtime”

Mentoring & Advising

2025	**Undergraduate students in Computer Science** Ethan Nicely and Rafael Dietsch JMU DIGITAL 2025
2024	**High School student** Ayush Jain High School Summer Internships/SPARKS 2024
2014	**Undergraduate student in Environmental Engineering (co-advisor)** Clarice Buarque de Macedo Lira Thesis: Analysis of the MM5 model ensemble forecasting system in the heavy rainfall event that occurred on April 5th and 6th, 2010 in Rio de Janeiro City

Talks and Presentations

C: conference/symposium; W: workshop, U: university

2024	**e-JUST: environmental justice using urban scalable toolkit**) Visiting lecture University of Illinois Chicago

2023	**Visual management and analysis of weather forecasts ensembles** Visiting lecture University of Illinois Chicago
2023	**Visual management and analysis of weather forecasts ensembles** Visiting lecture New York University
2022	**System to aid the creation, control, and visual analysis of numerical weather simulations** VI Climate Modeling Workshop (MODCLIM 6.0) Federal Fluminense University, Brazil
2021	**Perspectives for the evolution of atmospheric modeling at LAMMOC/UFF - high resolution MPAS and WRF** IV Climate Modeling Workshop (MODCLIM 4.0) Federal Fluminense University, Brazil

Course Development

2025	**Uncertainty Visualization** Conducted lecture and lab sessions for *Topics in Data Visualization* (Computer Science and Information Technology programs; approximately 10 junior and senior students) James Madison University
2025	**Data Visualization in Python** Conducted lecture and lab sessions for *Data Structures and Advanced Programming* (Information Technology program; approximately 20 freshmen and sophomore students) James Madison University
2021-2023	**Extension Course for Continued Education of Teachers in Financial Education for Sustainable Development** Multidisciplinary team member Federal Institute of Espírito Santo, Brazil

Services

Journal reviewer
Journal of the Brazilian Computer Society (2025)
Conference reviewer
IEEE PacificVis 2026 Papers; IEEE VIS 2025 Short Papers

Committees

Master's project committee
Leonamme Valvides da Costa Teixeira (IFES, 2025)
Undergraduate Final Project Course
Clarice Buarque de Macedo Lira (UFF, 2014)

Tools

ReactJS D3JS Deck.gl TypeScript Leaflet Mapbox.gl Vega-Lite Apache-Airflow FastAPI Flask NetCDF4 Matplotlib GeoPandas Pandas Xarray MonetDB MySQL Docker QGIS Linux Microsoft Pkgs

Languages

Portuguese (native)
English (advanced)
Spanish (basic)